Module 4 — Machine Learning Basics

Intern Works | Day 13–16 | High Priority Module

Module	Machine Learning Basics
Tools	Python, Scikit-learn, Pandas, Matplotlib
Tasks Covered	Student Score Prediction, House Price Prediction, Iris Flower Classification
Workflow	Data prep -> Train/Test Split -> Model Training -> Evaluation -> Visualization

This report documents three beginner-level supervised learning models built with Scikit-learn as part of Module 4. For each model: data was prepared, split into training and test sets, a model was trained, and performance was evaluated using appropriate metrics. Full source code, results, and visualizations are included below.

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1. Student Score Prediction

Type: Regression **Model:** Linear Regression **Goal:** Predict exam score from hours studied.

Code

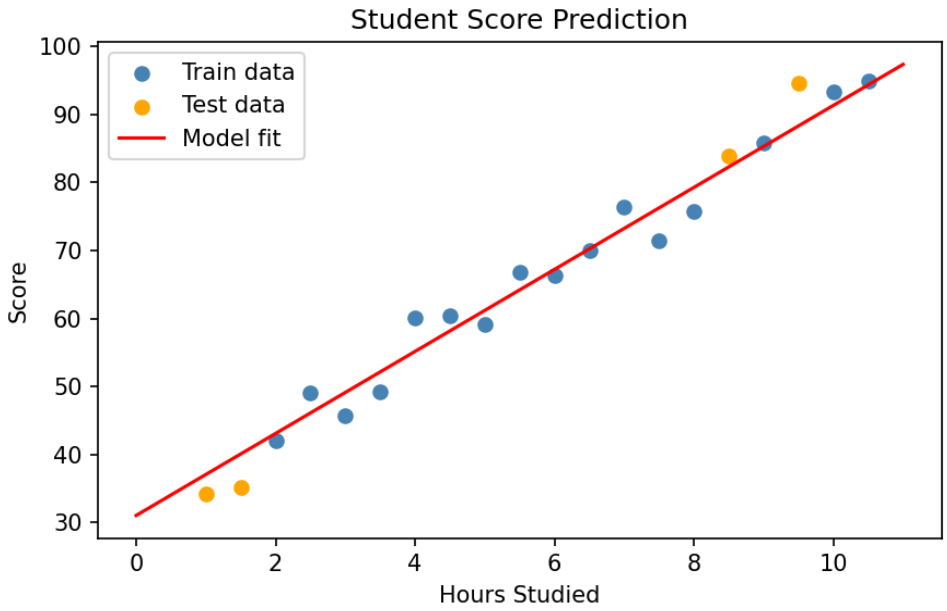
```
hours_studied = [...] # 20 data points
scores = 25 + 7.2 * hours_studied + noise

X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42)

model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Results

```
Training samples: 16  |  Test samples: 4
MAE: 3.94  |  RMSE: 4.33  |  R²: 0.975
Learned equation: score = 31.05 + 6.03 × hours_studied
```



The model explains 97.5% of the variance in test scores (R^2 = 0.975), indicating hours studied is a strong predictor of exam score in this dataset.

2. House Price Prediction

Type: Regression **Model:** Multi-feature Linear Regression **Goal:** Predict house price from size, bedrooms, and age.

Code

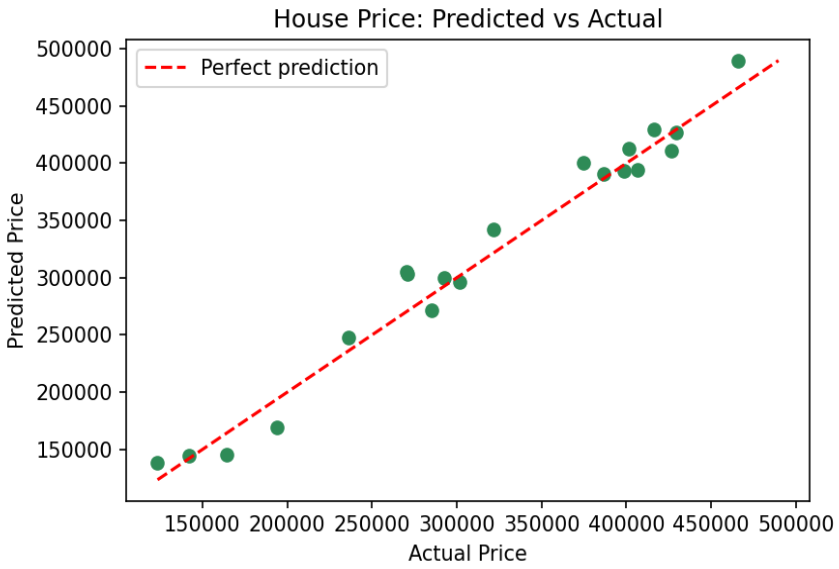
```
X = house_df[['size_sqft', 'bedrooms', 'age_years']]
y = house_df['price']

X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42)

model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Results

```
Training samples: 80  |  Test samples: 20
MAE: 14,920.91  |  RMSE: 17,606.80  |  R²: 0.970
Coefficients — size_sqft: 122.63, bedrooms: 7,569.76, age_years: -945.01
Intercept: 47,987.19
```



Predicted prices closely track actual prices (points near the red diagonal), with R^2 = 0.970. Size in sq ft is the dominant positive driver of price; age has a negative effect, as expected.

3. Iris Flower Classification

Type: Classification **Model:** Decision Tree Classifier **Goal:** Classify iris species from petal/sepal measurements.

Code

```
from sklearn.datasets import load_iris
iris = load_iris(as_frame=True)

X = iris.frame[iris.feature_names]
y = iris.frame['target']

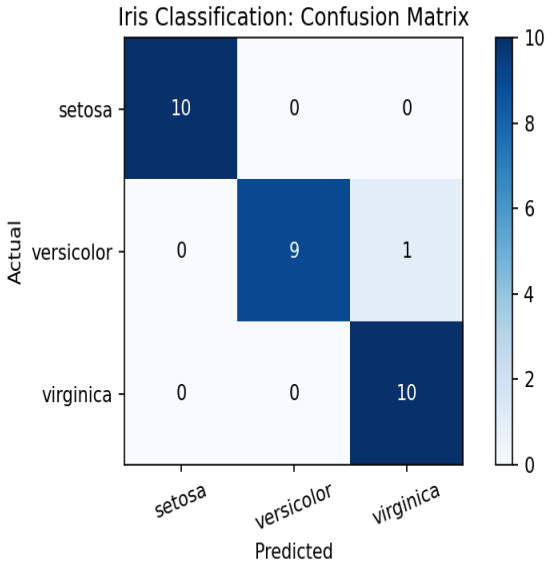
X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42, stratify=y)

model = DecisionTreeClassifier(max_depth=3, random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Results

```
Training samples: 120  |  Test samples: 30
Accuracy: 0.967 (29/30 correct)
```

Class	Precision	Recall	F1-score	Support
setosa	1.00	1.00	1.00	10
versicolor	1.00	0.90	0.95	10
virginica	0.91	1.00	0.95	10



The decision tree achieves 96.7% accuracy on the test set. The single misclassification occurs between versicolor and virginica — the two species with overlapping petal measurements — while setosa is perfectly separated.

Summary

Model	Type	Key Metric	Result
Student Score Prediction	Regression	R^2	0.975
House Price Prediction	Regression	R^2	0.970
Iris Flower Classification	Classification	Accuracy	96.7%

Key Learnings

- How to structure a complete Scikit-learn ML workflow: prepare data, split into train/test sets, fit a model, predict, and evaluate.
- The difference between regression metrics (MAE, RMSE, R^2) and classification metrics (accuracy, precision, recall, F1, confusion matrix).
- How model complexity and feature count affect performance and interpretability, comparing a single-feature model, a multi-feature model, and a tree-based classifier.

Prepared for Module 4 submission — Intern Works. Full runnable notebook available as the accompanying Google Colab link.